

30-Minute Student Task

Python Student Grade Management System — Solved

Name: K.Jayavardhanreddy

Register Number: 99230040324

Department: CSE

College: Kalasalingam University of Research and Education

1. Task Summary

The task is to build a simple Python-based Student Grade Management System. The program accepts student details and marks, calculates the total and average, assigns a grade based on the average, and displays the final result. The program also validates the marks so that invalid values are not accepted.

2. Requirement Checklist

Requirement	How it is satisfied
Accept student name	Uses input() to receive the student's name.
Accept subject marks	Uses input() and converts values to integers.
Validate marks	Checks that every mark is between 0 and 100.
Calculate total	Adds all subject marks.
Calculate average	Divides total marks by the number of subjects.
Assign grade	Uses if / elif / else based on average.
Display result	Prints total, average, and grade.

3. Complete Python Solution

```
name = input("Enter student name: ").strip()
marks = []

for i in range(3):
    while True:
        mark = int(input(f"Enter marks for Subject {i+1}: "))
        if 0 <= mark <= 100:
            marks.append(mark)
            break
        else:
            print("Enter marks between 0 and 100.")

total = sum(marks)
average = total / len(marks)

if average >= 90:
    grade = "A+"
elif average >= 80:
    grade = "A"
elif average >= 70:
    grade = "B"
elif average >= 60:
    grade = "C"
```

```
elif average >= 50:
    grade = "D"
else:
    grade = "F"

print("\nStudent Result")
print("Name:", name)
print("Total Marks:", total)
print("Average:", average)
print("Grade:", grade)
```

4. How the Program Works

INPUT: The program first accepts the student's name and marks for three subjects.

VALIDATION: Each mark is checked to make sure it is between 0 and 100.

PROCESS: The total and average marks are calculated using Python arithmetic operations.

GRADE: An if/elif/else structure assigns a grade according to the average.

OUTPUT: The student's name, total marks, average, and grade are displayed.

5. Test Cases and Expected Results

Test	Input	Expected Result
Student name	Rahul	Name is displayed correctly
High marks	95, 92, 90	Grade A+
Good marks	85, 82, 80	Grade A
Average marks	72, 70, 75	Grade B
Low marks	45, 50, 48	Grade F
Invalid marks	105	Program asks for valid marks

6. Sample Demonstration

Enter student name: Rahul

Enter marks for Subject 1: 85

Enter marks for Subject 2: 82

Enter marks for Subject 3: 88

Student Result

Name: Rahul

Total Marks: 255

Average: 85.0

Grade: A

7. Advantages of the System

The system is simple, easy to understand, and suitable for beginners learning Python. It demonstrates input handling, lists, loops, validation, arithmetic operations, and conditional statements. It also reduces manual calculation errors when determining student totals, averages, and grades.

8. Conclusion

The Python Student Grade Management System demonstrates how basic Python concepts can be combined to solve a practical problem. The program accepts student information, validates marks, calculates academic performance, assigns a suitable grade, and presents the result clearly. The system can be extended in the future to support multiple students, more subjects, grade reports, and file-based data storage.